SEMESTER: MAC – JULY 2024 (20242)

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TERM PAPER TITLE: **"AI CHATBOT AS A POWERFUL TOOL TO SUPPORT ACADEMIC RESEARCH"**

NAME OF CHATBOTS :

1. GEMINI AI
2. BLACKBOX AI

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1.0 INTRODUCTION

Artificial intelligence (AI) chatbots are smart virtual agents that engage in conversations with people. They use AI algorithms to understand and respond to natural language queries. These chatbots continuously improve their interactions by leveraging machine learning techniques, including natural language processing. They are deployed across various platforms, such as messaging apps and websites, and offer effective solutions for customer support, information retrieval, and automating tasks, leading to better user experiences and streamlined business processes.

Gemini AI represents an advanced artificial intelligence platform that revolutionizes how businesses assess and handle data. As a company specializing in predictive analytics, anomaly detection, and pattern recognition, Gemini AI leverages state-of-the-art machine learning algorithms to extract valuable insights from vast data sets. Their scalable solutions cater to various industries, including finance, healthcare, and e-commerce, integrating cutting-edge technologies like deep learning and neural networks. By doing so, they empower informed decision-making and enhance operational efficiency. Gemini AI equips companies with a versatile toolbox, enabling them to fully leverage their data and gain a competitive edge in today’s dynamic market landscape.

Blackbox AI refers to machine learning and artificial intelligence models that are intricate and opaque, making them difficult to understand or analyze. These models, often based on neural networks with multiple layers and complex algorithms, generate predictions or judgments. Despite their challenges, black box AI models find widespread use in various applications, including image and audio recognition, natural language processing, and decision-making systems. While these models can be highly accurate, caution is necessary during adoption and deployment due to their lack of transparency, which can lead to biases, unintended consequences, and errors.

2.0 GEMINI AI

At UiTM, the entrepreneurship program undergoes a transformative shift, embracing a holistic approach to fostering entrepreneurial mindsets and skills among students. The university establishes a dedicated entrepreneurship center equipped with state-of-the-art facilities and staffed with experienced mentors and industry experts. The curriculum is redesigned to incorporate interdisciplinary learning modules tailored to students' academic backgrounds and career aspirations, emphasizing practical experience, problem-solving, and innovation. Regular networking events, workshops, and guest lectures are organized. A comprehensive evaluation framework is implemented to track students' progress and measure the program's impact on both short-term skill development and long-term entrepreneurial success. Strategic partnerships are forged with external stakeholders, including businesses, government agencies, and alumni networks, enabling access to funding, expertise, and resources while fostering a culture of collaboration and innovation. As a result, UiTM's entrepreneurship program becomes a beacon of entrepreneurial excellence, empowering students to become visionary leaders and driving economic development in Malaysia and beyond.

Initial Prompt	Result
1. How can we ensure that the entrepreneurship program at UiTM effectively nurtures an entrepreneurial mindset among students, encouraging them to think creatively, take risks, and pursue innovative solutions to real-world challenges?	UiTM's entrepreneurship program can nurture an entrepreneurial mindset among students by focusing on a few key areas: **1. Experiential Learning:** • **Move beyond textbooks:** Integrate real-world case studies, guest lectures from successful entrepreneurs, and workshops on business simulations. • **Hands-on projects:** Incorporate projects where students develop business plans, pitch ideas, and get feedback from investors or industry professionals. • **Mentorship programs:** Connect students with experienced entrepreneurs who can provide guidance and support.

2. Fostering Creativity and Innovation:

- **Brainstorming and Design Thinking Workshops:** Teach students techniques to generate creative ideas and develop innovative solutions.
- **Encourage experimentation and calculated risk-taking:** Create a safe space for students to experiment with ideas and learn from failures.
- **Interdisciplinary courses:** Combine business courses with design thinking, engineering, or technology to encourage students to see problems from different angles.

3. Building Confidence and Risk Tolerance:

- **Success stories and role models:** Showcase successful alumni entrepreneurs to inspire students and demonstrate the possibilities.
- **Networking opportunities:** Organize events where students can connect with potential investors, mentors, and collaborators.
- **Entrepreneurship competitions:** Provide a platform for students to test their ideas and win seed funding or mentorship opportunities.

4. Resources and Support:

- **Incubation centers:** Offer physical workspaces and access to resources like software, legal advice, and prototyping facilities.
- **Funding opportunities:** Connect students with potential investors, grants, or crowdfunding platforms.
- **Mentorship network:** Provide ongoing support and guidance through a network of experienced entrepreneurs and business professionals.

By implementing these strategies, UiTM's entrepreneurship program can equip students with the skills, knowledge, and confidence to become successful entrepreneurs who tackle real-world challenges with innovative solutions.

Figure 2.1: Key Areas of Focus in UiTM's Entrepreneurship Program

Description of prompt:

The prompt outlines a plan for building a support system for entrepreneurs in UiTM’s program. The key areas of focus include mentorship, access to resources, and networking opportunities. Mentorship would involve matching students with mentors based on industry, experience, and student needs. There would also be workshops to train mentors on communication and coaching skills. Access to resources would include a physical or online resource center with business plan templates, market research tools, and financial planning resources. Students would also be given

access to relevant software programs and co-working spaces. Networking opportunities would include industry events, where students could connect with professionals and investors. There would also be alumni networks and pitch days where students could present their ideas and receive feedback.

Subsequence Prompt	Result
2. What strategies should be implemented to provide comprehensive support for students participating in the entrepreneurship program, including mentorship, access to resources, and opportunities for networking with industry professionals and investors?	▸ **Building a Support System for UiTM's Entrepreneurs** Here are some strategies to provide comprehensive support for students in UiTM's entrepreneurship program: **Mentorship:** • **Matching Program:** Develop a system to match students with mentors based on industry, experience, and student needs. • **Training for Mentors:** Provide workshops to equip mentors with effective communication and coaching skills. • **Mentorship Platforms:** Utilize online platforms to facilitate communication and scheduling between mentors and students. **Access to Resources:** • **Resource Center:** Create a physical or online resource center with access to business plan templates, market research tools, legal documents, and financial planning resources. • **Software and Technology Access:** Offer students access to relevant software programs, prototyping tools, and co-working spaces. • **Industry Partnerships:** Collaborate with local businesses to offer student discounts on software or services needed for venture development. **Networking Opportunities:** • **Industry Events:** Organize workshops, conferences, and guest speaker sessions featuring industry professionals and investors. • **Alumni Network:** Connect current students with alumni entrepreneurs for mentorship and networking opportunities. • **Pitch Days and Competitions:** Host events where students can present their ideas to potential investors and receive feedback.

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Additional Strategies:

Seed Funding Grants: Establish internal grant programs or partner with external organizations to provide seed funding for promising student ventures.
Entrepreneurship Workshops: Offer workshops on specific topics like business plan writing, financial modeling, pitching, and legal considerations for startups.
Career Services Collaboration: Partner with the university's career services department to offer guidance on entrepreneurial career paths and connect students with relevant job opportunities.

Tracking and Evaluation:

Track the progress of student ventures and measure the impact of the support system through surveys and focus groups.
Use this data to continuously improve the program offerings and ensure their effectiveness in nurturing a strong entrepreneurial ecosystem at UiTM.

By implementing these strategies, UiTM can create a comprehensive support system that empowers students to turn their innovative ideas into successful ventures.

Figure 2.2: Building a Support System for UiTM's Entrepreneurs
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<td colspan="2">
Description of prompt:

The prompt outlines a multi-faceted approach to nurturing entrepreneurial mindsets among students in UiTM’s program. It emphasizes experiential learning through real-world case studies, guest lectures, workshops, and hands-on projects. The program also focuses on fostering creativity and innovation through brainstorming sessions, encouraging calculated risks, and incorporating interdisciplinary courses. Additionally, it aims to build student confidence by showcasing successful alumni and providing networking opportunities, such as competitions and industry events.
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Subsequence Prompt	Result
3. How can we tailor the curriculum of the entrepreneurship program to address the specific needs and interests of UiTM students, considering factors such as their academic backgrounds, career aspirations, and cultural contexts?	Here's how UiTM's entrepreneurship program curriculum can be tailored to better address the needs and interests of its students: **Understanding the Student Body:** • **Conduct Needs Assessment Surveys:** Survey students to understand their academic backgrounds, career aspirations, and preferred learning styles. • **Focus Group Discussions:** Organize focus groups to gain deeper insights into students' challenges, interests, and cultural perspectives on entrepreneurship. • **Data Analysis of Student Demographics:** Analyze student data to identify subgroups with specific needs, like students from rural areas or those interested in social entrepreneurship. **Tailoring the Curriculum:** • **Modular Course Structure:** Develop a curriculum with core modules on business fundamentals and elective modules that cater to different interests, like social entrepreneurship, technological ventures, or Islamic finance principles. • **Real-world Case Studies:** Integrate case studies that resonate with the local context and feature successful Malaysian entrepreneurs. • **Incorporate Guest Lectures:** Invite guest speakers from relevant industries or alumni entrepreneurs to share their experiences and challenges faced in the Malaysian market. • **Culturally Sensitive Approach:** Frame concepts and discussions within a framework that acknowledges and respects the cultural values and norms of Malaysian society. **Additional Considerations:** • **Language Options:** Offer course materials or lectures in Bahasa Malaysia alongside English to cater to students' language preferences. • **Incorporate Islamic Business Practices:** Discuss Islamic financing options, ethical considerations, and how they can be integrated into business models. • **Entrepreneurship in the Digital Age:** Focus on equipping students with the skills necessary to leverage digital technologies for business development and marketing in the Malaysian context. **Flexibility and Support:** • **Mentorship Programs:** Match students with mentors who understand their cultural background and can provide guidance tailored to their specific goals. • **Industry Mentorship Opportunities:** Connect students with industry professionals who can offer insights into specific sectors relevant to their ventures. • **Career Counseling:** Offer career counseling services to help students explore different entrepreneurial career paths and navigate the startup ecosystem in Malaysia. By tailoring the curriculum and incorporating these additional considerations, UiTM's entrepreneurship program can become more relevant and engaging for its students, empowering them to become successful entrepreneurs within the Malaysian context.

<table>
<tr><td></td><td>Figure 2.3: Tailoring the UiTM Entrepreneurship Curriculum for Student Success</td></tr>
<tr><td colspan="2">Description of prompt:

The prompt outlines a curriculum for an entrepreneurship program focused on developing key skills in students. It emphasizes experiential learning through real-world case studies, guest lectures, and hands-on projects. The program also aims to foster creativity and innovation through brainstorming and encouraging calculated risks. It highlights the importance of building student confidence by showcasing successful alumni and providing opportunities to network with industry professionals.</td></tr>
</table>

Subsequence Prompt	Result
4. What measures can be implemented to evaluate the success and impact of the entrepreneurship program, both in terms of short-term outcomes such as skills development and long-term outcomes such as the establishment of successful startups and contributions to economic development?	Evaluating the success of UiTM's entrepreneurship program requires a multi-faceted approach that considers both short-term and long-term outcomes. Here are some key measures to consider: **Short-Term Outcomes (Skills Development):** • **Pre- and Post-Program Assessments:** Administer standardized tests or surveys to measure student knowledge gain in key areas like business planning, finance, and marketing before and after program completion. • **Project-Based Learning Evaluation:** Assess the quality of student projects, business plans, and pitches to gauge their understanding of core entrepreneurial concepts and their ability to apply them. • **Student Surveys and Focus Groups:** Gather student feedback on their learning experience, skill development, and satisfaction with the program's effectiveness in equipping them with the necessary skills. • **Mentorship Program Evaluation:** Collect feedback from mentors and students on the effectiveness of the mentorship program in fostering skill development and providing guidance.

Long-Term Outcomes (Startups and Economic Impact):

- **Startup Formation Rates:** Track the number of startups launched by program alumni within a specific timeframe (e.g., 2-5 years after graduation).
- **Startup Growth and Revenue Generation:** Monitor the revenue and growth of alumni-founded startups to assess their economic impact.
- **Job Creation:** Track the number of jobs created by alumni-founded startups, contributing to economic development.
- **Innovation and Social Impact:** Evaluate the degree to which alumni startups are developing innovative solutions and addressing social challenges.

Additional Considerations:

- **Longitudinal Studies:** Conduct longitudinal studies that track program alumni over time to gain a deeper understanding of their career paths, entrepreneurial success, and long-term impact.
- **Employer Surveys:** Survey employers who hire graduates of the program to assess their satisfaction with the graduates' entrepreneurial skills and preparedness.
- **Community Engagement:** Track the program's contribution to the local entrepreneurial ecosystem through participation in startup events, community mentorship initiatives, or partnerships with incubators and accelerators.

Data Collection and Analysis:

- Utilize a combination of quantitative data (surveys, assessments, startup data) and qualitative data (interviews, focus groups) to get a holistic view of the program's impact.
- Establish a system for collecting and analyzing data over time to track progress and identify areas for improvement.

By implementing these measures and continuously evaluating the program's effectiveness, UiTM can ensure that its entrepreneurship program is equipping students with the necessary skills and knowledge to not only develop their ventures but also contribute meaningfully to the economic and social development of Malaysia.

Figure 2.4: Fostering Creativity and Confidence at UiTM

Description of prompt:

The prompt outlines a comprehensive entrepreneurship program designed to cultivate well-rounded future business owners. The program emphasizes experiential learning through real-world case studies, workshops, and mentorship opportunities. It also fosters creativity and innovation through brainstorming techniques and encourages calculated risk-taking. To build student confidence, the program showcases successful alumni and provides networking opportunities with industry professionals.

Finally, it equips students with the resources they need to succeed through facilities, funding opportunities, and ongoing mentorship.

Subsequence Prompt	Result
5. How can partnerships with external stakeholders, including businesses, government agencies, and alumni networks, be leveraged to enhance the effectiveness and sustainability of the entrepreneurship program at UiTM, fostering greater collaboration, funding opportunities, and access to expertise and resources?	UiTM's entrepreneurship program can leverage partnerships with external stakeholders in several ways to boost its effectiveness, sustainability, and overall impact: **Collaboration:** • **Industry Mentorship Programs:** Partner with businesses to connect students with experienced professionals who can provide real-world guidance and industry insights. • **Project-based Learning:** Collaborate with businesses on real-world projects, allowing students to apply their knowledge and gain practical experience. • **Guest Speaker Series:** Invite entrepreneurs, investors, and industry leaders to share their expertise and inspire students through lectures and workshops. • **Startup Competitions:** Organize competitions with industry partners, offering students the chance to win seed funding, mentorship opportunities, or incubation space. **Funding Opportunities:** • **Venture Capital Partnerships:** Partner with venture capital firms or angel investor networks to connect promising student ventures with potential funding sources. • **Grant Proposals:** Collaborate with government agencies or foundations to secure grant funding for specific program initiatives or student ventures with social impact. • **Fundraising Events:** Organize fundraising events with alumni and industry partners to raise capital for program resources and student scholarships. **Access to Expertise and Resources:** • **Industry Advisory Board:** Establish an advisory board with representatives from businesses, government agencies, and alumni to provide strategic direction and ensure the program stays relevant to industry needs. • **Software and Technology Access:** Partner with tech companies to offer students access to relevant software, prototyping tools, or cloud computing resources at discounted rates. • **Incubation Space:** Collaborate with businesses or government agencies to establish an incubation center offering workspaces, mentorship, and access to essential resources for student startups.

Leveraging Alumni Networks:

- **Mentorship Programs:** Connect alumni entrepreneurs with current students as mentors, sharing their experiences and offering valuable career guidance.
- **Angel Investor Network:** Encourage successful alumni to invest in student ventures, providing seed funding and fostering a culture of giving back.
- **Industry Networking Events:** Organize events where students can connect with alumni working in various industries, exploring job opportunities or potential collaborations.

Sustainability Measures:

- **Develop clear partnership agreements** outlining expectations, responsibilities, and benefits for all parties involved.
- **Maintain open communication channels** with stakeholders to ensure successful collaboration and program alignment with industry needs.
- **Showcase the program's impact** through success stories of alumni ventures and data on job creation and economic contributions.

By fostering strong partnerships with external stakeholders, UiTM's entrepreneurship program can create a vibrant ecosystem that empowers students with the knowledge, skills, resources, and networks they need to launch and grow successful ventures, ultimately contributing to a thriving entrepreneurial landscape in Malaysia.

Figure 2.5: Maximizing External Partnerships for Enhanced Entrepreneurship Program Effectiveness at UiTM

Description of prompt:

The prompt outlines a comprehensive UiTM's entrepreneurship program to enhance its impact by engaging in partnerships with businesses, government agencies, and alumni. Through collaboration with industry, students can benefit from mentoring, project-based learning, and access to experts. These partnerships also provide funding opportunities via venture capitalists, grant proposals, and fundraising events. Industry advisory boards and alumni networks contribute strategic guidance and potential seed funding for student ventures, ultimately fostering student success and contributing to Malaysia’s economic growth.

2.1 Academic Benefit of Gemini AI

a. Advantages
   - Gemini Ai provides **personalized recommendations** based on student specific prompts to know about particular entrepreneurship information programs in the university, which provide specific responses that allow students to understand what needs to be done in entrepreneurship programs.
   - Gemini AI provides quick collective essential information regarding entrepreneurship programmes, which **saves students time and effort** in research, resulting in students working efficiently.
   - Gemini AI provides access to **extensive knowledge** bases on relevant topics and information of entrepreneurship programs, which helps students explore a wide range of issues and give them a better understanding.

b. Limitation
   - Gemini AI limits students' **ability to track and check the accuracy of the information** presented since it does not include the references and links that students need to ensure that the information is reliable.
   - Gemini AI's suggestions for customizing entrepreneurship programmes might be based on **old data and patterns** since it doesn't get real-time feedback on how students' interests and needs change. Therefore, the suggestions provided might not be tailored to the current trend.
   - Gemini AI may **not fully grasp the diverse backgrounds, goals, and learning styles** of UiTM students, leading to some students being disengaged with its entrepreneurial advice. By gathering more data on individual interests and aspirations, Gemini AI can personalize recommendations to better cater to each student's unique needs and enhance their learning experience.

3.0 BLACKBOX AI

In this scenario, BLACKBOX AI will be used to implement the create, read, and update functionality for booking appointments. Firstly, to create user-friendly interfaces where users can input their appointment details starts with the selecting date and time to make a reservation for the appointment then their name, identification card number and address. Next, users should be able to easily access and view their upcoming appointments. Additionally, it's crucial to incorporate reminder features, ensuring users are notified of their scheduled appointments. This includes QR codes for identification purposes, and confirming that the patient attends the appointment as scheduled, based on the booked date and time. Finally, the system enables automatic updating of existing appointments as users attend them. If a user fails to attend, provide a notification offering them the option to reschedule the appointment.

<table>
<tr><th>Initial Prompt</th><th>Result</th></tr>
<tr><td>

1. Create a form where users can choose the date and time for the appointment, after that input details such as name, contact information, and any additional notes. Upon submission, validate the input data and add the appointment request to the database.

</td><td>

```html
<form action="/appointment" method="post">
  <label for="appointment_date"> Date and Time for Appointment:</label>
  <input type="datetime-local" id="appointment_date"
name="appointment_date" required><br><br>

  <label for="name">Name:</label>
  <input type="text" id="name" name="name" required><br><br>

  <label for="contact">Contact Information:</label>
  <input type="text" id="contact" name="contact" required><br><br>

  <label for="notes">Additional Notes:</label>
  <textarea id="notes" name="notes"></textarea><br><br>

  <input type="submit" value="Submit Appointment Request">
</form>
```

Figure 3.1: HTML code of appointment form.

</td></tr>
</table>

```javascript
const form = document.querySelector('form');

form.addEventListener('submit', async (e) => {
  e.preventDefault();

  const appointmentDate = document.querySelector('#appointment_date').value;
  const name = document.querySelector('#name').value;
  const contact = document.querySelector('#contact').value;
  const notes = document.querySelector('#notes').value;

  if (!appointmentDate ||!name ||!contact) {
    alert('Please fill in all required fields');
    return;
  }

  try {
    const appointment = new Appointment({
      appointmentDate,
      name,
      contact,
      notes
    });

    await appointment.save();

    alert('Appointment request submitted successfully!');
  } catch (error) {
    console.error(error);
    alert('Error submitting appointment request');
  }
});
```

Figure 3.2: JavaScript code of appointment form.

Description of prompt:

After sending the initial prompt, the results showed the HTML code of an appointment form for the user to select the date and time, along with details such as name, contact information, additional notes, and a submit button. Then, the BLACKBOX AI also provided JavaScript to validate the data provided in the appointment form filled by the user. As the form requires all fields to be filled, it will display a popup message if the user fails to do so. Additionally, there is a popup notification to inform the user if the appointment was successfully submitted or if there was an error while submitting the appointment form.

<table>
<tr><th>Subsequence Prompt</th><th>Result</th></tr>
<tr><td>

2. After submitting the form for the appointment, create the access button to make it easy for the user to easily access the upcoming appointment that has been booked. Also, if the date and time are already booked, create a message for the user to book another date and time.

</td><td>

```html
<form action="/appointment" method="post">
  <!-- form fields here -->
  <input type="submit" value="Submit Appointment Request">

  <!-- access button will be displayed here after submission -->
  <div id="access-button" style="display: none;">
    <button>Access Upcoming Appointment</button>
  </div>

  <!-- error message will be displayed here if date and time are already
  <div id="error-message" style="display: none; color: red;">
    Sorry, the date and time you selected are already booked.
    Please choose another date and time.
  </div>
</form>
```

Figure 3.3: HTML code after submitting the appointment form.

```javascript
const form = document.querySelector('form');
const accessButton = document.querySelector('#access-button');
const errorMessage = document.querySelector('#error-message');

form.addEventListener('submit', async (e) => {
  e.preventDefault();

  // validate form data and add appointment request to database
  try {
    // if successful, display access button and hide error message
    accessButton.style.display = 'block';
    errorMessage.style.display = 'none';
  } catch (error) {
    // if date and time are already booked, display error message
        and hide access button
    errorMessage.style.display = 'block';
    accessButton.style.display = 'none';
  }
});
```

Figure 3.4: JavaScript of the validation for date and time.

</td></tr>
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Description of prompt:

The next prompt has a message indicating that the user has successfully booked the appointment. As the prompt requests a button to view the appointments that have been booked, it should be added accordingly to make the user easily access the submitted form. Additionally, if the date and time are already booked, a message should be created for the user to select another date and time.

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<table>
<tr><th>Subsequence Prompt</th><th>Result</th></tr>
<tr><td>

3. Add the reminder features to ensure the user notified the upcoming appointment as scheduled.

</td><td>

```javascript
// Store the appointment details in a database
const appointment = {
  name: 'User Name',
  contact: 'user@example.com',
  date: '2024-05-14T10:00:00',
  notes: 'Additional notes here',
};

// Set reminder time for 1 hour from now
const reminderTime = Date.now() + 3600000;

// Schedule the reminder
setTimeout(() => {
  const appointmentFromDatabase = getAppointmentFromDatabase();

  // Retrieve the appointment details from the database
  sendNotification(appointmentFromDatabase);
}, reminderTime - Date.now());

// Example function to send a notification via email
function sendNotification(appointment) {
  const message = `Hello ${appointment.name}, \n\n
This is a reminder that you have an
appointment scheduled for ${appointment.date}. \n\n${appointment.notes}
\n\nBest regards, \nYour Appointment Reminder System`;

  // Send the email using a service like SendGrid or SMTP
  sendEmail(appointment.contact, 'Appointment Reminder', message);
}
```

Figure 3.5: JavaScript of reminder features for upcoming appointment

</td></tr>
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Description of prompt:

This prompt is to add a reminder feature to the system. The purpose of this reminder feature is to ensure that the user is informed of their upcoming appointment as per the scheduled time. It works in a system that will send a notification or alert to the user, reminding them of their appointment before it happens. This notification is via means such as text message notification.

</td></tr>
</table>

Subsequence Prompt	Result
4. The appointment form includes a QR code for identification purposes, confirming that the patient attends the appointment as scheduled according to the booked date and time.	

```javascript
const QRCode = require('qrcode');

// Example appointment details
const appointment = {
  name: 'John Doe',
  date: '2024-05-14T10:00:00',
  notes: 'Additional notes here'
};

// Generate a QR code for the appointment
QRCode.toDataURL(JSON.stringify(appointment), (err, dataUrl) => {
  if (err) {
    console.error(err);
    return;
  }

  // The dataUrl variable now contains the QR code as a data URL
  // You can include this data URL in the appointment form
});
```

Figure 3.6: JavaScript of appointment form includes a QR code

Description of prompt:

This prompt describes the appointment form feature, which is the document or interface used to schedule an appointment. The feature described here is the entry of a QR code on the appointment form. The QR code serves for identification purposes. When the patient attends the appointment as scheduled, the QR code can be scanned or read by the device to confirm their attendance. This verification process ensures that the patient arrives at the appointment on the booked date and time.

<table>
<tr><th>Subsequence Prompt</th><th>Result</th></tr>
<tr><td>5. Add the if-else case in the code to enable automatic updating of existing appointments as users attend the appointment. If a user fails to attend, provide a notification giving them the option to reschedule the appointment.</td><td>

```
function attendAppointment(appointmentId, userId) {
  appointment = getAppointmentById(appointmentId);

  if (appointment.attendanceStatus == "attended") {
    // User has already attended the appointment
    return;
  }

  // Update the user's attendance status
  updateAttendanceStatus(appointmentId, userId, "attended");

  // Check if the user failed to attend the appointment
  if (appointment.attendanceStatus == "missed") {
    // Send a notification to the user
    sendRescheduleNotification(appointmentId, userId);
  }
}
```

Figure 3.7: JavaScript of automatic updating of existing appointments</td></tr>
<tr><td colspan="2">Description of prompt:

This prompt instructions entails modifying or adding functionality to a piece of code to facilitate automatic updating of existing appointments based on user attendance. Firstly, incorporate an if-else statement into the code, a conditional statement in programming, to execute different blocks of code based on certain conditions. The purpose of this addition is to automatically update appointment records when users attend their appointments. If a user fails to attend, the code should generate a notification, providing them with the option to reschedule the missed appointment.</td></tr>
</table>

3.1 Academic Benefit of BLACKBOX AI

a. Advantages

- Blackbox AI **improves students' software problem-solving abilities** for scenarios such as appointment scheduling. It customizes code snippets in many languages based on user prompts to meet individual needs and delivers prompt feedback, which students can learn and develop skills from the responses.
- Blackbox AI allows students to **easily access and refer to some credible sources** of knowledge at the beginning of AI responses that are related to booking appointments. The sources given would provide additional help in doing the student's work, which guarantees that the student's work is thoroughly researched and supported by reputable sources, thus improving the overall quality and trustworthiness of the Blackbox AI responses.
- In the context of black box AI systems, users are provided with a **friendly interface and pre-configured operations**, which are usable by students with varying levels of technical expertise. This ease of use makes it possible for students to practice AI tools.

b. Limitation

- Blackbox AI solutions **rely on pre-trained models** that may not be suitable for student projects or educational goals. This lack of customization and adaptability can make it difficult for students to find solutions to possible coding problems of booking appointments.
- Students' engagement with Blackbox AI systems **limits their opportunity for hands-on learning and experimentation** compared to other educational methods. Without understanding the inner workings of algorithms and processes, students may struggle to deepen their knowledge and progress.

- Blackbox AI systems could **hinder students' creativity** by providing fixed answers. Lack of freedom to explore new approaches about booking appointments may impede opportunities for advancements in AI.

4.0 CONCLUSION

The exploration of Gemini AI and Blackbox AI in the context of supporting academic research has provided valuable insights into the potential benefits and limitations of leveraging artificial intelligence for university students.

Gemini AI offers personalized recommendations and quick access to essential information, enhancing students' efficiency in research and exploration of entrepreneurship programs. However, its limitations include the inability to verify the accuracy of presented information and the potential reliance on outdated data, which may hinder its effectiveness in catering to evolving student needs.

On the other hand, Blackbox AI demonstrates the ability to improve students' problem-solving skills in computer science and provide access to credible sources of knowledge. Its user-friendly interface and pre-configured operations make it accessible to students in coding and do as references for projects or revision. However, the reliance on pre-trained models limits customization and may hinder hands-on learning and creativity.

In conclusion, while both Gemini AI and Blackbox AI offer unique advantages in supporting academic research and learning, there is a need for careful consideration of their limitations. Integrating these AI tools into educational environments requires a balance between efficiency and fostering critical thinking, creativity, and hands-on learning experiences for students. Moreover, ongoing efforts are necessary to address the challenges posed by AI in education and ensure its effective integration to enhance student learning outcomes and academic research endeavors.

5.0 REFERENCES

Ivankov, A. (2024, February 8). *Advantages and Disadvantages of Google Gemini - Profolus*. Profolus. https://www.profolus.com/topics/advantages-disadvantages-bard/

Soffar, H. (2024, May 2). *Gemini AI review, advantages, disadvantages and Is Gemini AI better than ChatGPT?* | Science online. *Science online*. https://www.online-sciences.com/robotics/gemini-ai-review-advantages-disadvantages-and-is-gemini-ai-better-than-chatgpt/

Tyagi, A. (2024, April 25). *How to solve all your Coding Problems using Blackbox AI?* Analytics Vidhya. https://www.analyticsvidhya.com/blog/2024/02/how-blackbox-ai-can-solve-all-your-coding-problems/